

Learning & Remembering

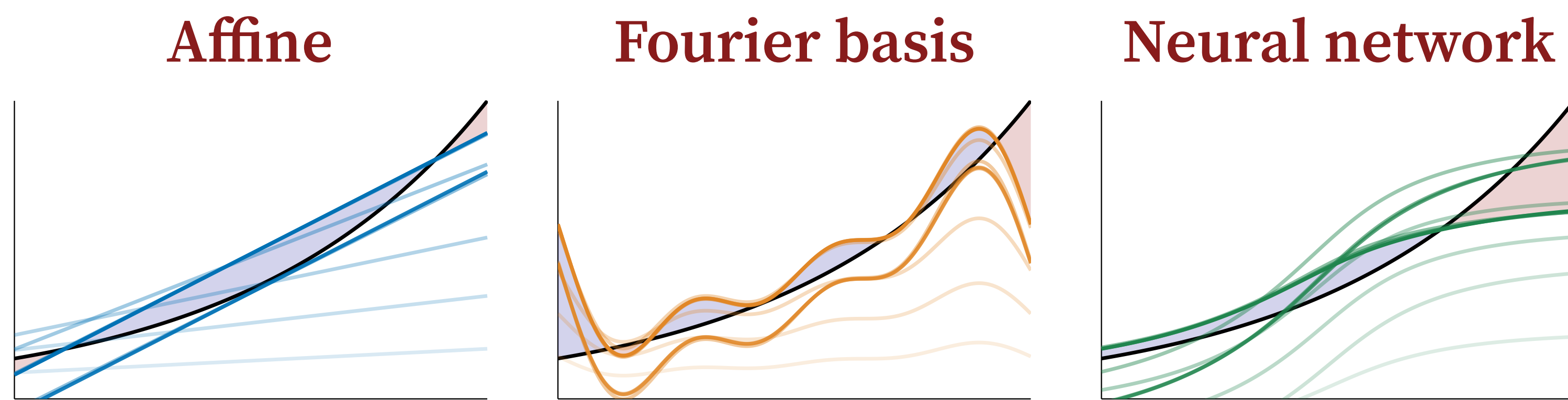
Problems are about action. Learning should be too.

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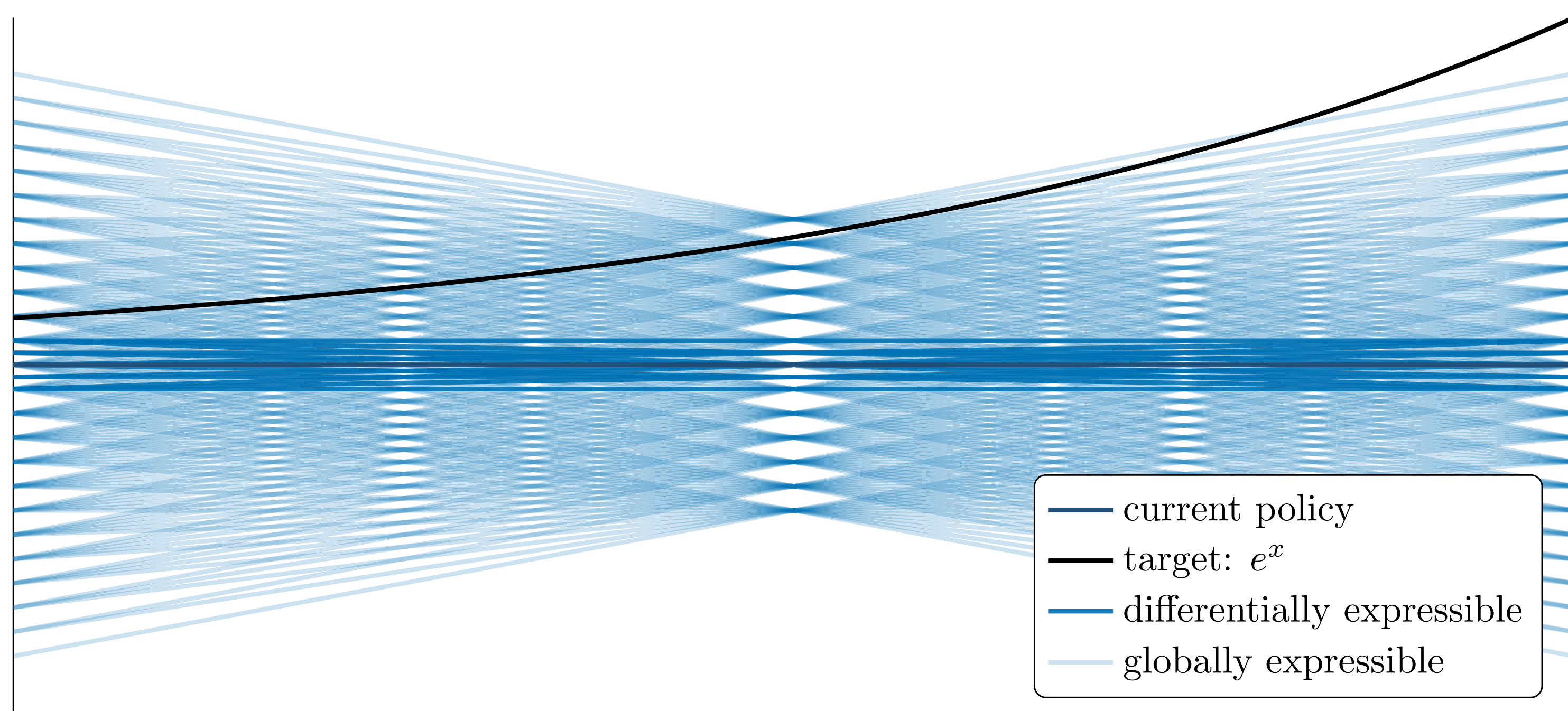
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What would you do?

Learning is change in a policy, $f_{t+1} - f_t$. Learning rules determine what “should” be learned. What can be learned is restricted by the structure of the agent’s brain.



The functions f_θ can express



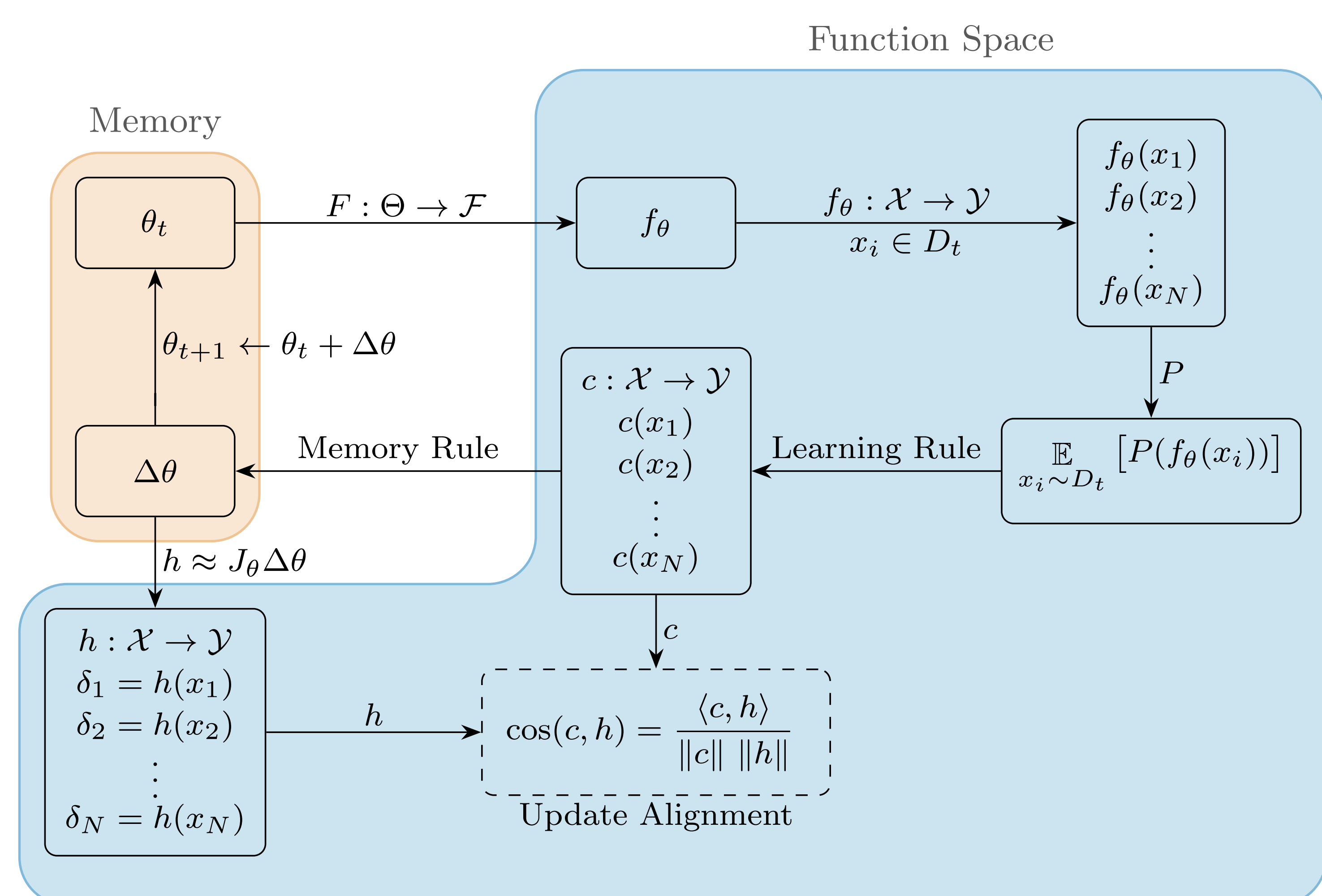
Global expressivity is what an architecture could express with *any* parameters. It says nothing about what a learner, carefully adjusting its parameters, can reach.

Learning is richer than parameter change

Feedback from the problem is in terms of situations and actions—not parameters.

Learning signals agents determine how to change their policies. These decisions are functions $c : \mathcal{X} \rightarrow \mathcal{Y}$, like policy f and the functional gradient $\nabla_f P$.

Memory is the representation of the policy. The memory rule receives signal c and updates parameters by $\Delta\theta$, but the resulting update h may not be aligned with c .



Performance improvement factors

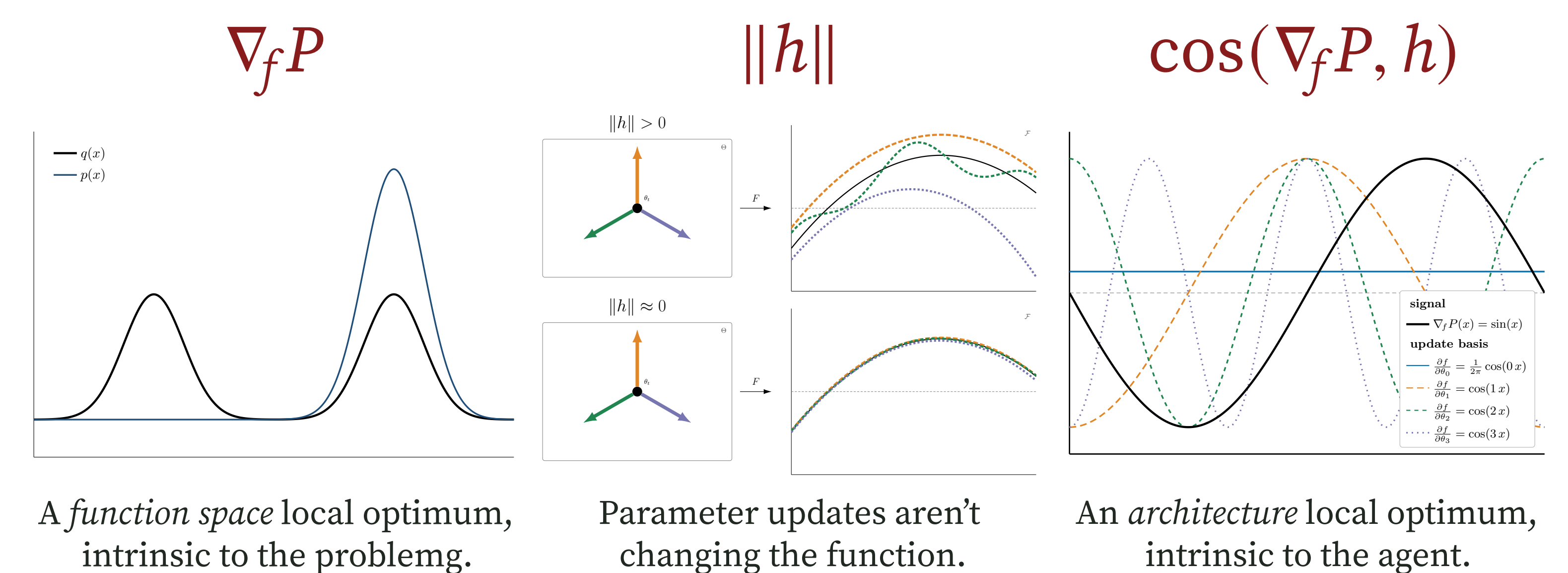
Improvement is, to first order, a product of three terms:

$$\Delta P = \underbrace{\cos(\nabla_f P, h)}_{\text{alignment}} \times \underbrace{\|h\|}_{\text{magnitude}} \times \underbrace{\|\nabla_f P\|}_{\text{signal}} + O(\|h\|^2)$$

The agent controls magnitude and alignment, but not the gradient. When step size is limited, **alignment is the only thing left to improve**.

Why isn't your agent learning?

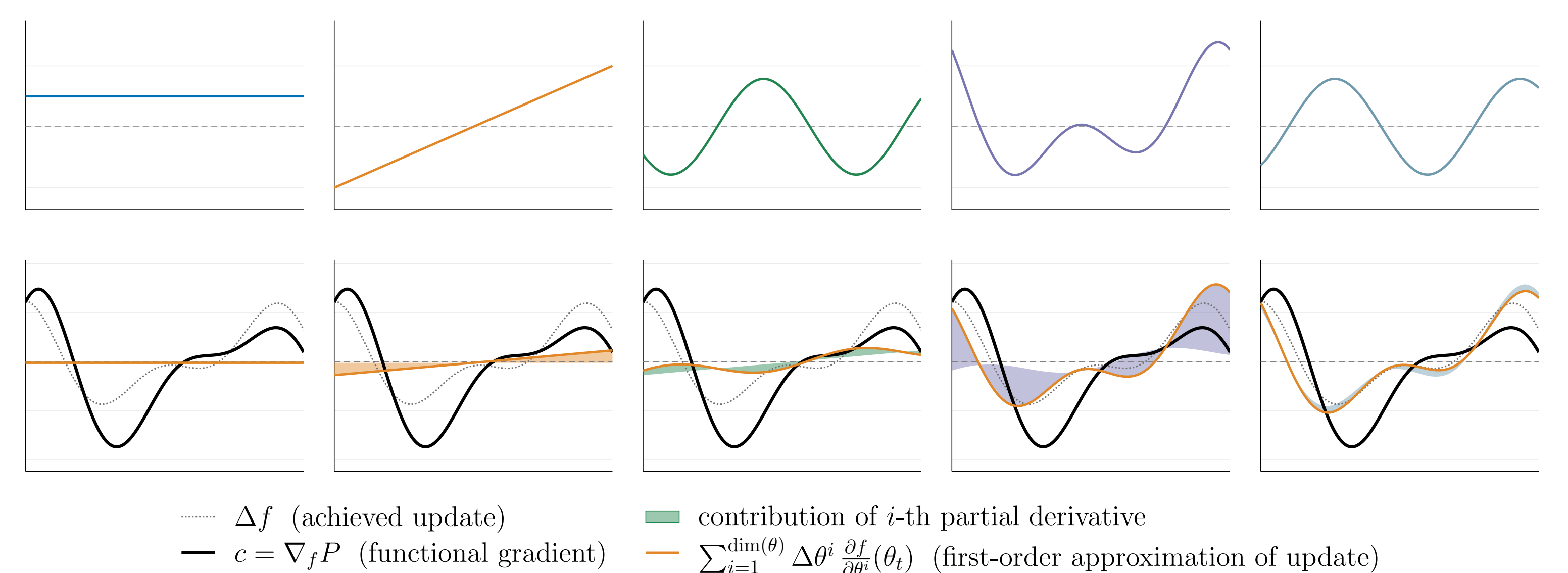
When performance isn't improving, $P(f_{t+1}) \leq P(f_t)$, there's a problem with one of the three factors—but they're entangled as a “local optimum” in parameter space:



$f_{\theta_{t+1}}$ must be in the differential expressivity

Small parameter changes lead to policy updates in the architecture's tangent space:

$$f_{\theta_{t+1}} \approx f_{\theta_t} + \sum_{i=1}^{\dim(\theta)} \Delta\theta^i \frac{\partial f}{\partial \theta^i}(\theta_t)$$



Differential expressivity bounds alignment; $\cos(c, h) \leq \sup_g \cos(c, g)$ over the updates g the agent can express. Natural gradients make the most of it, but differential expressivity can be poorly aligned.

Directed plasticity

Definition. The alignment between the signal c and the update h ;

$$\cos(c, h)$$

Improvement = alignment × magnitude × signal. Architecture constrains alignment. Choose accordingly.

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Large language models were used throughout this project.

